

Notes on lecture 10

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1 Method of Lagrange multipliers

1.1 For a function of 3 variables

Let f be a function of 3 variables x, y, z . If we want to extremize the value of f subject to some constraint $g(x, y, z) = k$, then as before (in the case of functions of 2 variables), we have to draw the level surfaces of f . We will get infinitely many level surfaces. One of these level surfaces (say, $f(x, y, z) = c_1$) will touch the surface $g(x, y, z) = k$. These point(s) of touch are the points of extremum of f . At these point(s), if we draw the tangent planes to these surfaces, then those tangent planes must coincide. This means that the normals to these tangent planes at those point(s) must be parallel to each other. That is, if (x_0, y_0, z_0) is such a point, then we must have:

$$\nabla f(x_0, y_0, z_0) = \lambda \nabla g(x_0, y_0, z_0) \text{ for some scalar } \lambda \neq 0.$$

The scalar λ is called a **Lagrange multiplier**.

Example 1.1.1. Let us redo Example 3.0.1 of the last lecture using the method of Lagrange multipliers.

Here the volume $V(x, y, z) = xyz$ and $g(x, y, z) = xy + 2xz + 2yz$. We have to maximize $V(x, y, z)$ subject to $g(x, y, z) = 12$. Let (x_0, y_0, z_0) be a point of maximum. Now

$$\nabla V(x_0, y_0, z_0) = \lambda \nabla g(x_0, y_0, z_0)$$

$$\Rightarrow \langle y_0 z_0, x_0 z_0, x_0 y_0 \rangle = \lambda \langle y_0 + 2z_0, x_0 + 2z_0, 2x_0 + 2y_0 \rangle.$$

$$\Rightarrow \underline{y_0 z_0 = \lambda(y_0 + 2z_0)}, \quad \underline{z_0 x_0 = \lambda(x_0 + 2z_0)}, \quad \underline{x_0 y_0 = \lambda(2x_0 + 2y_0)}. \quad (+)$$

Also, we have the condition that

$$\underline{x_0 y_0 + 2x_0 z_0 + 2y_0 z_0 = 12.} \quad (*)$$

From the equations (+), we can write

$$x_0 y_0 z_0 = \lambda(x_0 y_0 + 2z_0 x_0)$$

$$x_0 y_0 z_0 = \lambda(x_0 y_0 + 2z_0 y_0)$$

$$x_0 y_0 z_0 = \lambda(2x_0 z_0 + 2y_0 z_0).$$

Since none of x_0, y_0, z_0 is equal to 0 (because they represent length, breadth, and height) it follows from the first two equations above that

$$2z_0 x_0 = 2z_0 y_0 \Rightarrow x_0 = y_0.$$

Similarly, from the second and third equations, it follows that

$$x_0 y_0 = 2x_0 z_0 \Rightarrow y_0 = 2z_0 \Rightarrow z_0 = \frac{y_0}{2}.$$

Again from (*), we have

$$x_0 x_0 + 2x_0 \frac{x_0}{2} + 2x_0 \frac{x_0}{2} = 12$$

$$\Rightarrow x_0 = \pm 2.$$

Now x_0 represents length, so it cannot be negative. So we have

$$x_0 = y_0 = 2, \quad z_0 = 1.$$

Therefore the maximum value of $V(x, y, z) = V(x_0, y_0, z_0) = x_0 y_0 z_0 = 2 \cdot 2 \cdot 1 = 4$ meter cubes. $\square$

2 The method of Lagrange multipliers for more than one constraints

Suppose we are trying to maximize/minimize $f(x, y, z)$ subject to 2 constraints

$$g(x, y, z) = k$$

and

$$h(x, y, z) = c.$$

We know that the 2 given constraints are nothing but level surfaces of the 2 functions g and h . The intersection of these 2 level surfaces is going to be

a curve, say C . We are seeking the maximum/minimum values of $f(x, y, z)$ along the curve C . Say, at a point $P(x_0, y_0, z_0)$ of the curve C , we are going to get a maximum/minimum value of f . Consider a tangent line to C at the point P . This tangent line must lie on the tangent planes to the surfaces $g(x, y, z) = k$ and $h(x, y, z) = c$ both. We have to consider a third surface corresponding to $f(x, y, z)$ in such a way that the curve C touches the surface $f(x, y, z) = \text{constant}$ at the point P . So ∇f is orthogonal to C at P . Figure 2.0.1 explains this phenomenon.

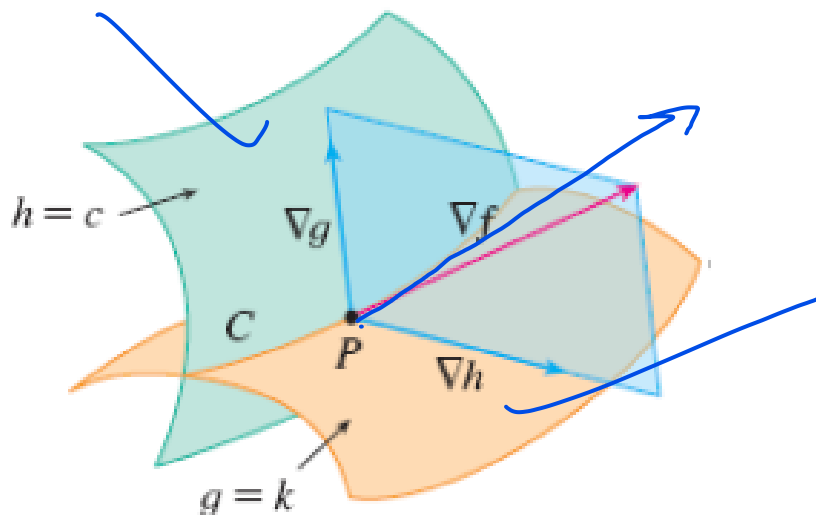


Figure 2.0.1: The method of Lagrange multipliers

We also know that ∇g is orthogonal to $g(x, y, z) = k$ and ∇h is orthogonal to $h(x, y, z) = c$. So ∇g and ∇h are both orthogonal to C . This means that the gradient vector $\nabla f(x_0, y_0, z_0)$ lies in the plane determined by $\nabla g(x_0, y_0, z_0)$ and $\nabla h(x_0, y_0, z_0)$ (We assume here that these gradient vectors are not zero and not parallel). Please see Figure 2.0.2 below.

That means, the normal $\nabla f(x_0, y_0, z_0)$ must be a linear combination of the other 2 normals at the point P . That is,

$$\nabla f(x_0, y_0, z_0) = \lambda \nabla g(x_0, y_0, z_0) + \mu \nabla h(x_0, y_0, z_0),$$

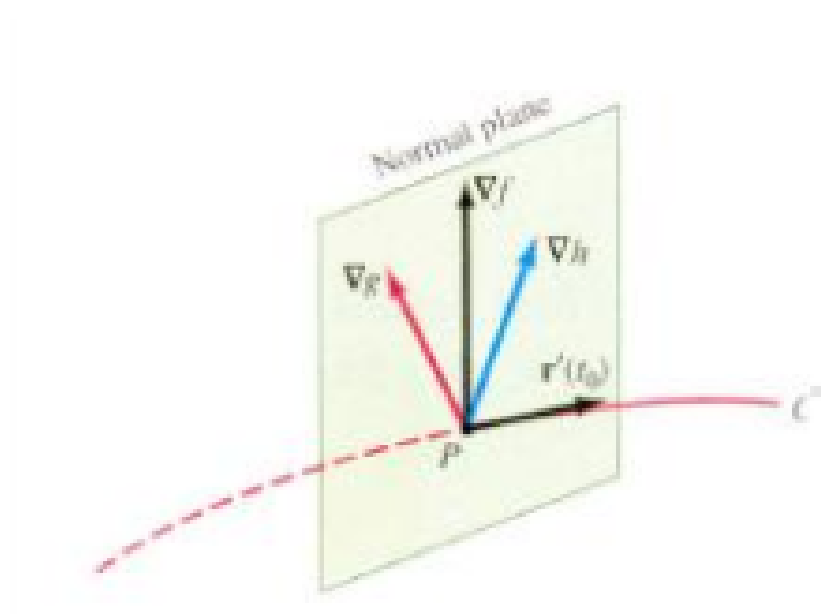


Figure 2.0.2: The 3 normals in the same plane

for some scalars λ and μ . These scalars λ and μ are called **Lagrange multipliers**.

Example 2.0.1. Find the maximum value of the function $f(x, y, z) = x + 2y + 3z$ on the curve of intersection of the plane $x - y + z = 1$ and the cylinder $x^2 + y^2 = 1$. See Figure 2.0.3 below. Let $g(x, y, z) = x - y + z$ and

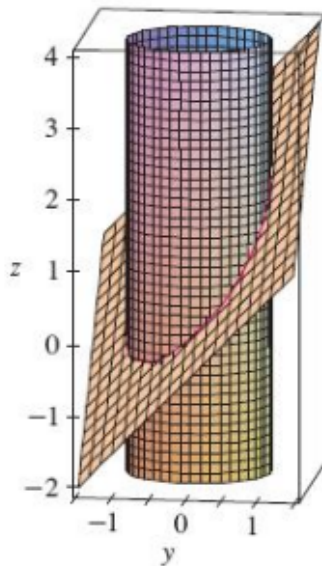


Figure 2.0.3: Figure for Example 2.0.1

$h(x, y, z) = x^2 + y^2$. Now $\nabla f = \lambda \nabla g + \mu \nabla h$ implies

$$\langle 1, 2, 3 \rangle = \langle \lambda + 2\mu x, -\lambda + 2\mu y, \lambda \rangle.$$

This gives

$$\lambda + 2\mu x = 1$$

$$-\lambda + 2\mu y = 2$$

$$\lambda = 3.$$

Other than these 3 relations, we also have the relations

$$x - y + z = 1$$

and

$$x^2 + y^2 = 1.$$

Substituting $\lambda = 3$ in the first equation, we get $2\mu x = -2$, which gives $x = -\frac{1}{\mu}$. Similarly, the second equation gives us $y = \frac{5}{2\mu}$. Plugging in $x = -\frac{1}{\mu}$ and $y = \frac{5}{2\mu}$ in $x^2 + y^2 = 1$, we get

$$\frac{1}{\mu^2} + \frac{25}{4\mu^2} = 1,$$

which implies that $\mu = \pm \frac{\sqrt{29}}{2}$.

When $\mu = \frac{\sqrt{29}}{2}$, we have $x = -\frac{2}{\sqrt{29}}$, $y = \frac{5}{\sqrt{29}}$, and the value of z and then be calculated easily using the relation $x - y + z = 1$. Similarly, we can find out x, y, z when $\mu = -\frac{\sqrt{29}}{2}$.

For these 2 sets of values of x, y, z , we can easily compute the values of f . We will get 2 values of f in this process. Whichever value of f is greater (among these 2 values), will be the maximum value of f . $\square$

3 Areas of surfaces of revolution

Consider the x -axis. Suppose we have a curve $y = f(x)$ in the xy -plane. Suppose we are rotating this curve about the x -axis. Then it will generate a surface. In this section, we will find out the area of this generated surface. In order to do this, we will need some concepts from single variable calculus. Let us first discuss those concepts.

3.1 Back to single variable calculus

If we want to find out the length of a curve, which is given as the graph of a function of a single variable, then we divide the interval on which the curve is defined into certain subintervals, and then through different points lying on the curve (corresponding to the end points of those subintervals), we construct a polygon. Please see Figure 3.1.1 below.

After that, we let the number of subintervals tend to infinity. We can observe in Figure 3.1.2 that as the number of subintervals increase, the polygon approximation to the curve becomes better and better.

Then the sum of the lengths of all the line segments on the polygon is the

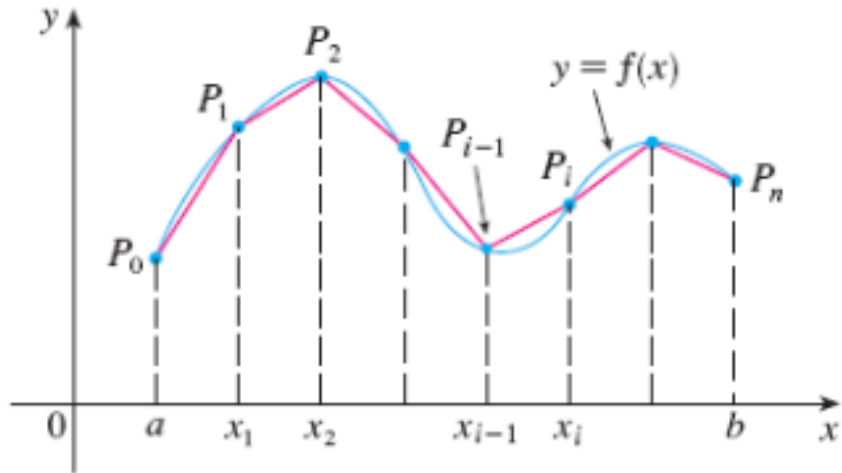


Figure 3.1.1: Polygon approximating a curve

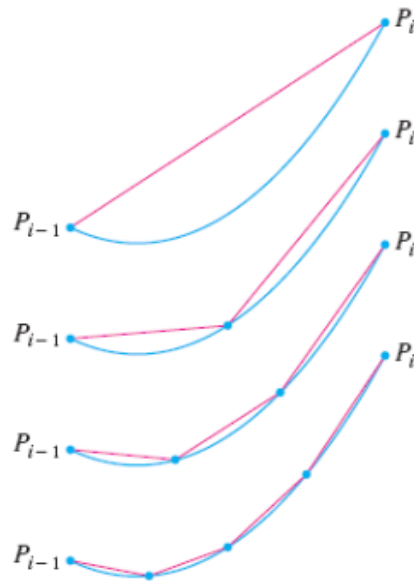


Figure 3.1.2: Number of subintervals increasing

length L of the curve. That is,

$$L = \lim_{n \rightarrow \infty} \sum_{i=1}^n |P_{i-1}P_i|.$$

Recall the **arc length formula** from one of the previous lectures: If f' is continuous on $[a, b]$, then the length of the curve $y = f(x)$, $a \leq x \leq b$, is

$$L = \int_a^b \sqrt{1 + [f'(x)]^2} dx.$$

This can be further written as

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx.$$

If we consider the curve as a function of y , that is, if we consider x as the dependent variable and y as the independent variable, and the curve has equation $x = g(y)$, $c \leq y \leq d$, and $g'(y)$ is continuous, then by interchanging the roles of x and y (and replacing $f(x)$ by $g(y)$) in the above two equations, we obtain

$$L = \int_c^d \sqrt{1 + [g'(y)]^2} dy = \int_c^d \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy.$$

The arc length function: If we consider the curve from the initial point a upto a variable point x , then the arc length function is given by

$$s(x) = \int_a^x \sqrt{1 + [f'(t)]^2} dt,$$

where t is a dummy variable. Differentiating the above equation, we get

$$\frac{ds}{dx} = \sqrt{1 + [f'(x)]^2} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}.$$

The above equation can also be written in symmetric form as

$$(ds)^2 = (dx)^2 + (dy)^2.$$

Geometrical interpretation: Consider the curve given in Figure 3.1.3 below. Consider a tangent to the curve at the point $(x, f(x))$. Let $dx = \Delta x$

and $\Delta y = f(x + \Delta x) - f(x)$. Let dy be the distance between $f(x)$ and the point on the tangent line corresponding to the point $x + \Delta x$ on the x -axis. From the figure, one can easily see that $(ds)^2 = (dx)^2 + (dy)^2$. This is the geometrical interpretation.

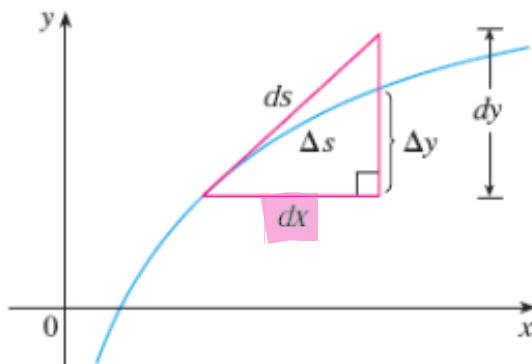


Figure 3.1.3: Geometrical interpretation: The arc length function

3.2 Some basic and necessary calculations

Suppose we consider a circular cylinder of height h and radius r . If we cut the surface of this cylinder vertically, and exfoliate it, we will get a rectangle. Figure 3.2.1 below illustrates the phenomenon. Then the breadth of the

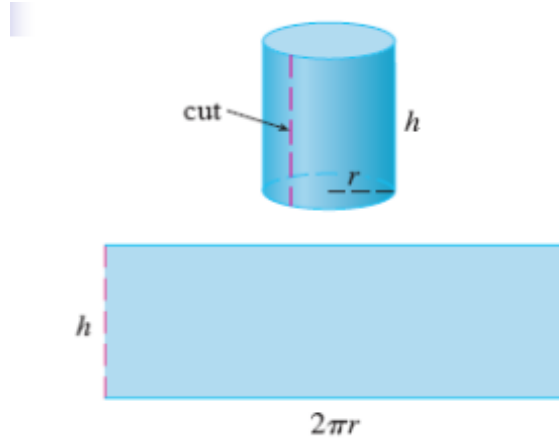


Figure 3.2.1: A circular cylinder cut into a rectangle

rectangle will be h and length will be $2\pi r$. So the area of the rectangle will be equal to $2\pi rh$.

Now, consider a cone, whose length of the slanted side is l and the radius of the base is r . Suppose we cut the surface of the cone from the slanted side and exfoliate it. Then we are going to get a shape of a polar triangle having angle $\theta = \frac{2\pi r}{l}$. Please see Figure 3.2.2 for an illustration. We know from elementary calculus that the area of this polar triangle equals $\frac{1}{2}l^2\theta = \frac{1}{2}l^2\frac{2\pi r}{l} = \pi rl$.

Suppose now, we want to find out the surface area of a cone upto a certain height from the bottom (This portion of the cone looks like the shade of a table lamp). Say, the radius of the lower base of the portion of the cone is r_2 , and that of the upper base is r_1 . Say, the length of the slanted side is l . If we extend this structure to a full cone, then the area of the earlier portion (which looks like the shade of a table lamp) equals

$$\text{Area of the big cone} - \text{Area of the smaller cone.}$$

Let l_1 denote the length of the slanted side of the smaller cone. Please see Figure 3.2.3 for an illustration.

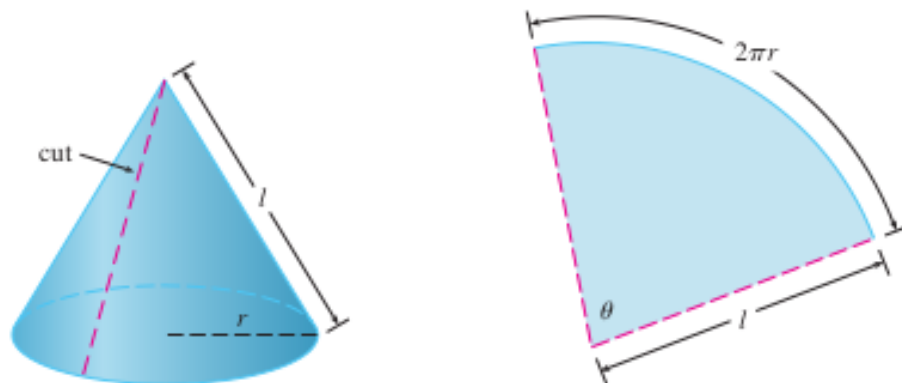


Figure 3.2.2: A cone cut into a polar triangle

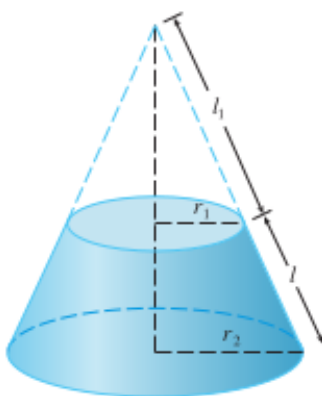


Figure 3.2.3: A portion of a cone

Then we have:

Area of the portion looking like the shade of a table lamp

= Area of the big cone – Area of the smaller cone

$$= \pi r_2(l_1 + l) - \pi r_1 l_1.$$

Consider the 2 right angled triangles formed by the bigger and the smaller cones. These 2 triangles are similar. This implies that

$$\frac{r_2}{r_1} = \frac{l_1 + l}{l_1}$$

$$\Rightarrow (r_2 - r_1)l_1 = r_1 l.$$

Hence we have

Area of the portion looking like the shade of a table lamp

$$= \pi r_2(l_1 + l) - \pi r_1 l_1$$

$$= \pi(r_2 - r_1)l_1 + \pi r_2 l = \pi r_2 l + \pi r_1 l = \pi(r_1 + r_2)l.$$

So the area of the surface of the portion looking like the shade of a table lamp equals $\pi(r_1 + r_2)l$.

3.3 Back to area of surface of revolution

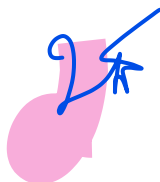
Suppose we are considering the xy -plane and in that plane, we are considering the graph of a function f from a to b . Assume that f is continuous and differentiable on $[a, b]$. We are seeking the area of the surface of revolution of the curve $y = f(x)$ ($a \leq x \leq b$) about the x -axis.

We subdivide the interval $[a, b]$ by the points $a = x_0, x_1, \dots, x_{n-1}, x_n = b$. Let us now focus on one subinterval $[x_{i-1}, x_i]$. Suppose the points on the curve $y = f(x)$ corresponding to x_{i-1} and x_i are P_{i-1} and P_i respectively. Let us join P_{i-1} and P_i by a straight line. Then we are going to get a “band” like structure (as shown in red color in part (b) of Figure 3.3.1 below). We are going to find out the area of this band. Observe that this band looks like the “table lamp-shaped” portion of a cone, whose area we had calculated in the previous subsection.

For each such subinterval $[x_{i-1}, x_i]$, we are going to get such a band. We are going to sum up the areas of all such bands and then let the number of subintervals tend to infinity. In this way, we will get the entire area of the surface of revolution.

The area of the band equals $\pi|P_{i-1}P_i|(y_i + y_{i-1})$, where $y_i = f(x_i)$ and $y_{i-1} = f(x_{i-1})$. Therefore, the area of the surface of revolution equals

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \pi(y_{i+1} + y_i)|P_{i-1}P_i|.$$



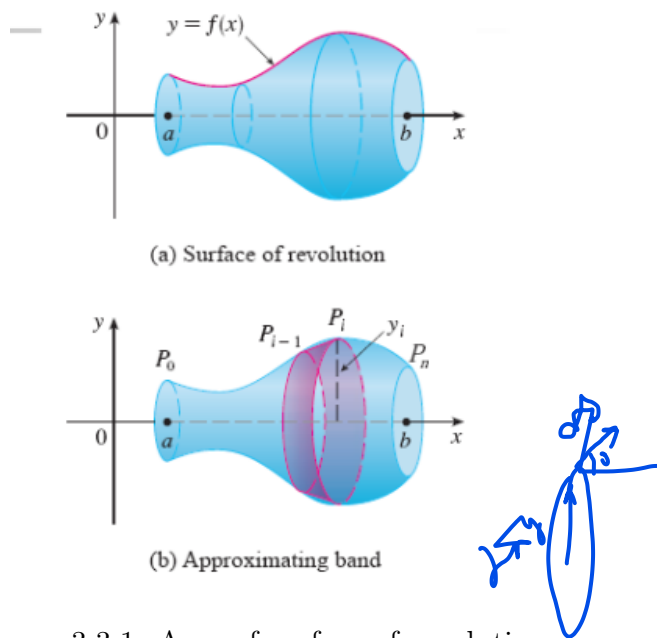


Figure 3.3.1: Area of surface of revolution

Recall that

$$|P_{i-1}P_i| = \sqrt{1 + [f'(x_i^*)]^2} \Delta x_i,$$

where $\Delta x_i = x_i - x_{i-1}$ and $x_i^* \in (x_{i-1}, x_i)$. So the area of the surface of revolution

$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n \pi(y_{i-1} + y_i) \sqrt{1 + [f'(x_i^*)]^2} \Delta x_i.$$

Note that the above sum is a Riemann sum. In this sum, y_{i-1} can be substituted with y_i and x_i^* can be substituted with x_i . So the above expression equals

$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n 2\pi y_i \sqrt{1 + [f'(x_i)]^2} \Delta x_i = \int_a^b 2\pi y \sqrt{1 + [f'(x)]^2} dx.$$

Remark 3.3.1. The above concept is based on the fact that the curve $y = f(x)$ always lies above the x -axis. That is, $f(x) \geq 0$ for all $x \in [a, b]$.

Therefore, we have:

If f is positive and has a continuous derivative, then

S = Surface area of revolution about the x - axis

$$= \int_a^b 2\pi f(x) \sqrt{1 + [f'(x)]^2} dx. \quad (1)$$

With the Leibniz notation for derivatives, the above formula becomes

$$S = \int_a^b 2\pi y \sqrt{1 + \left[\frac{dy}{dx}\right]^2} dx. \quad (2)$$

The function $y = f(x)$ can also sometimes (not always) be expressed as $x = g(y)$. So if the curve is described as $x = g(y)$, $c \leq y \leq d$, then the formula for the surface area becomes

$$S = \int_c^d 2\pi y \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy. \quad (3)$$

Note here that the circumference is not going to change. So the $2\pi y$ -part of the formula in equation (2) will remain as it is in equation (3). The axis of revolution is the x -axis, that is why there is a $2\pi y$ in the formula (3), NOT $2\pi x$. The part $\sqrt{1 + \left(\frac{dy}{dx}\right)^2}$ in equation (2) was coming from the distance formula, that's why in equation (3), the roles of x and y got interchanged. The formulae (2) and (3) can be summarized as

$$S = \int 2\pi y ds,$$

where the limits of integration will be decided depending upon the fact that we are differentiating with respect to x or y .

For revolution about the y -axis, the surface area formula becomes

$$S = \int 2\pi x ds.$$

Note here that for revolution about the y -axis, the circumference formula is going to involve x , because we are going to take distance from the y -axis, and this distance we are going to measure along the x -axis.

Example 3.3.2. The curve $y = \sqrt{4 - x^2}$, $-1 \leq x \leq 1$, is an arc of the circle $x^2 + y^2 = 4$. Find the area of the surface obtained by revolving this arc about the x -axis. The surface is a portion of a sphere of radius 2. Please see Figure 3.3.2 below.

By formula (2), the required area equals $\int_{-1}^1 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$. Here

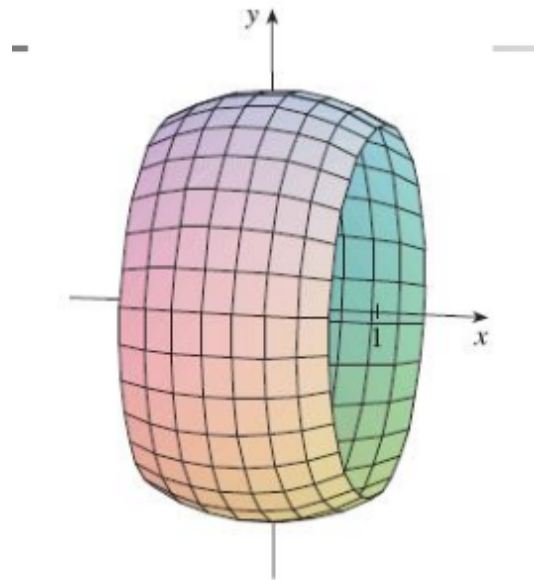


Figure 3.3.2: Figure for Example 3.3.2

$$\frac{dy}{dx} = -\frac{x}{\sqrt{4-x^2}}.$$

So the required area

$$= \int_{-1}^1 2\pi\sqrt{4-x^2} \frac{2}{\sqrt{4-x^2}} dx = \int_{-1}^1 4\pi dx = 8\pi.$$

By formula (3), the required area

$$\begin{aligned} &= 2 \int_{\sqrt{3}}^2 2\pi y \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy \\ &= 2 \int_{\sqrt{3}}^2 2\pi y \frac{2}{\sqrt{4-y^2}} dy. \end{aligned}$$

Please think why we have multiplied the integral $\int_{\sqrt{3}}^2 2\pi y \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$ with the integer 2.

Check that the required area comes out to be 8π in this case as well. $\square$

Remark 3.3.3. To calculate some area of a surface of revolution, we can use any formula. Whichever formula is simpler, we should go for that.

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